SOVEREIGN GOVERNMENT CLOUD

CSIR Secure Web Appliance OS (CSIR-SWA OS)

Next-Generation Hardened, Immutable, and Containerized Web Hosting Appliance

CORE ARCHITECTURAL MANDATE

- Engineered for classified research laboratories, government web portals, and critical public infrastructure.
- Immutable openSUSE MicroOS transactional engine with zero-malware persistence.
- Integrated multi-tenant Web Hosting, Virtual Host Management, Reverse Proxy, and ACME SSL Key Vault.

STRATEGIC IMPERATIVES

Institutional Objectives & Sovereign Control

Challenges in Government Web Hosting

- **Commercial Vendor Lock-in:** Heavy recurring licensing costs for enterprise operating systems, commercial WAFs, and load balancers.
- **High Vulnerability Surface:** Standard mutable Linux distros allow unauthorized file tampering, backdoor persistence, and malware footholds.
- **Compliance Bottlenecks:** Complex CERT-In 6-hour incident disclosure rules require tedious manual log parsing.

The CSIR-SWA OS Solution

- **Zero Licensing Overhead:** 100% open-source sovereign software stack saves millions in proprietary software budgets.
- **Immutable System Integrity:** Read-only Btrfs root (/) ensures that unauthorized changes cannot persist across reboots.
- **One-Click CERT-In Compliance:** Append-only SHA-256 cryptographic audit ledger guarantees 180-day retention and certified audit export.

CORE ARCHITECTURE

Layered Sovereign Defense Architecture

System Foundation & Isolation

- **Layer 1:** Bare-Metal Hardware with TPM 2.0 cryptographic attestation.
- **Layer 2:** openSUSE MicroOS Immutable Core with Transactional Update engine.
- **Layer 3:** Btrfs subvolumes (@, @/var, @/srv) providing isolated atomic snapshots.
- **Layer 4:** Rootless Podman Quadlets running with dropped Linux capabilities.

Edge Defense & Operations

- **Layer 5:** nftables default-deny ingress packet filtering and Indian NPL Stratum 1 NTP sync.
- **Layer 6:** Coraza Sovereign WAF with OWASP Core Rule Set (CRS 4.x) inspection.
- **Layer 7:** HAProxy L7 Dynamic Reverse Proxy & Virtual Host Management.
- **Layer 8:** Local AI Security Center providing zero-cloud threat correlation and ChatOps.

NATIVE WEB HOSTING

Multi-Tenant Virtual Host & Web Serving Subsystem

Comprehensive Virtual Host Features

- **Multi-Tenant Isolation:** Independent document roots allocated under /srv/www/vhosts//.
- **Multi-Runtime Support:** Native execution pools for PHP 8.2-FPM, Node.js 20 SSR, Python ASGI (Uvicorn), and static web assets.
- **High-Performance Protocols:** Strict HTTP/2 and modern HTTP/3 (QUIC) multiplexing enabled by default.
- **Quota & Resource Governance:** Per-vhost storage quotas, bandwidth caps, and segregated access/error logs.

Sovereign Web Serving Benefits

- **Eliminates dependency on third-party commercial control panels (cPanel/Plesk).:**
- **Full integration with systemd socket activation for on-demand worker spin-up.:**
- **Automated subvolume snapshot binding per virtual host for instantaneous rollback.:**
- **Direct inspection by Coraza WAF prior to reaching any web worker processes.:**

TRAFFIC ORCHESTRATION

Dynamic Reverse Proxy & Layer 7 Load Balancer

Upstream Management & Algorithms

- **HAProxy Sovereign Router:** Sub-millisecond latency edge ingress and path-based routing.
- **Dynamic Load Balancing:** Round-Robin, Weighted Round-Robin, Least Connections, and IP-Hash session pinning.
- **Granular Path Matching:** Route rules such as /api/v1/* to API clusters and /ingest/* to high-rate sensor nodes.
- **Deep Health Probing:** Active HTTP/TCP health checks with configurable fail counters and automated failover.

Operational Resilience & Scaling

- **Circuit Breaker Engine:** Automatically trips failing backends to prevent cascading cluster outages.
- **Transparent Streaming:** Native support for long-lived WebSockets, HTTP/2 Server Push, and gRPC channels.
- **Header Security Injection:** Automated enforcement of HSTS, CSP, X-Frame-Options, and Referrer-Policy.
- **Zero Downtime Reloads:** Graceful state transfer during upstream pool reconfiguration.

CRYPTOGRAPHIC PROTECTION

Sovereign SSL/TLS Certificate Vault & Key Management

Automated Certificate Lifecycle

- **ACME Let's Encrypt Integration:** Fully automated issuance and 30-day pre-expiry renewal via HTTP-01/DNS-01.
- **Wildcard & Multi-SAN Support:** Seamless deployment of *.serc.res.in and research domain certificates.
- **Custom Sovereign Key Import:** Secure vault for institutional certificates and HSM-backed private keys.
- **CSR Generator:** In-browser generation of RFC-compliant CSRs with SHA-384 / ECDSA secp384r1 curves.

Hardened Cipher Suites & CERT-In Standards

- **Strict TLS 1.3 Mandate:** Deprecated protocols (TLS 1.0, 1.1, and legacy ciphers) are strictly disabled.
- **OCSP Stapling:** High-speed certificate revocation checking with zero DNS/privacy leakage.
- **Tamper-Evident Key Vault:** Encrypted storage with rootless access controls and TPM endorsement keys.
- **Cryptographic Compliance:** 100% compliant with Indian National Cyber Security Directives.

WORKLOAD ISOLATION

Rootless Podman Quadlets & Systemd Integration

Daemonless System Architecture

- **Systemd-Native Quadlets:** Workloads defined as declarative .container and .network systemd unit files.
- **Rootless Security:** Containers execute entirely inside unprivileged user namespaces (UserNS).
- **Dropped Capabilities:** High-risk Linux privileges (CAP_SYS_ADMIN, CAP_NET_ADMIN) dropped by default.
- **Read-Only Root Filesystems:** Container roots are mounted read-only, preventing runtime infection.

Cryptographic Image Provenance

- **Cosign Digital Signatures:** Only images signed by trusted CSIR institutional keys are permitted to boot.
- **Digest Pinning:** All container images are strictly bound by immutable SHA-256 digests.
- **SBOM Validation:** Automated validation of Software Bill of Materials against CVE databases.
- **Hard Resource Quotas:** Deterministic CPU, RAM, and I/O limits prevent noisy-neighbor starvation.

PERIMETER DEFENSE

Coraza Sovereign WAF & nftables Network Shield

Real-Time Threat Neutralization

- **OWASP Core Rule Set (CRS 4.x):** Deep packet inspection for SQL Injection, XSS, Remote Code Execution, and LFI.
- **Zero False-Positive Engine:** Dynamic anomaly scoring with adjustable paranoia levels (PL1 – PL4).
- **Ingress Rate Limiting:** High-frequency token bucket limiting prevents brute-force credential stuffing.
- **Bot & Scanner Neutralization:** Automated drop policies for illicit vulnerability scanners and crawlers.

Automated Threat Escalation

- **nftables Integration:** WAF block events immediately escalate offensive subnets into kernel packet drop sets.
- **24-Hour Adaptive Isolation:** Repeat offenders quarantined without incurring user-space CPU overhead.
- **Complete Sovereign Telemetry:** Full visibility into rule triggers, target endpoints, and payload decodes.
- **Zero Third-Party Data Leakage:** All inspection is 100% on-premises without external telemetry phone-home.

AUTONOMOUS OPERATIONS

Local AI Security & Operations Center (Zero Cloud)

Autonomous Threat Narrative Synthesis

- **Air-Gapped Local LLM:** Security telemetry analyzed locally without transmitting data to external clouds.
- **Plain-English Threat Narratives:** Converts thousands of raw WAF logs into clear executive incident briefings.
- **Root Cause Analysis (RCA):** Autonomous diagnostics pinpoint application database bottlenecks and query contention.
- **Capacity Telemetry:** Proactive forecasting of storage subvolume consumption and compute spikes.

Sovereign Human Governance Gate

- **Strict Advisory Role:** AI can propose remediation actions but can NEVER autonomously execute high-impact commands.
- **Human Authorization Barrier:** Critical patches, firewall modifications, and rollbacks require verified admin sign-off.
- **Integrated ChatOps Assistant:** Natural-language command interface for telemetry querying and maintenance staging.
- **Immutable Audit Trail:** All AI recommendations and human decisions are permanently recorded in the SHA-256 ledger.

BUSINESS CONTINUITY

Data Protection, Btrfs Snapshots & DR Drills

Subvolume Tree Protection

- **Copy-on-Write Snapshots:** Zero-overhead instantaneous Btrfs snapshots across @, @/var, and @/srv.
- **Automated Snapper Timeline:** Hourly, daily, and pre-update snapshots retained according to policy.
- **Encrypted Restic Repositories:** Off-site, deduplicated, and cryptographically encrypted secondary backups.
- **Ransomware Immunity:** Read-only snapshot trees prevent crypto-locker manipulation or deletion.

Automated DR Verification Drills

- **Scheduled Sandbox Drills:** Automated recovery verification tests executed in isolated network namespaces.
- **Verified RTO & RPO:** Exact measurement of Recovery Time Objective (12s – 23s) and Recovery Point Objective.
- **Cryptographic Integrity Checks:** Full SHA-256 verification of restored database tables and application datasets.
- **Audit-Ready Certification:** Automated generation of DR readiness scorecards for audit inspections.

REGULATORY COMPLIANCE

CERT-In Section 70B Cryptographic Audit Ledger

Regulatory Mandate Compliance

- **Mandatory 180-Day Retention:** Immutable system logs locked against premature truncation or alteration.
- **Directions under Section 70B:** Comprehensive tracking of cybersecurity incidents, system changes, and access attempts.
- **Indian NPL Stratum 1 Clock Sync:** Strict microsecond clock discipline synchronized directly to National Physical Laboratory Delhi.
- **Cryptographic Hash Chaining:** Every audit record contains SHA256(prev_hash | payload_hash | timestamp | actor).

Rapid Incident Disclosure

- **6-Hour Statutory Window:** One-click export of certified incident dossiers formatted for CERT-In reporting.
- **Tamper-Evident Ledger:** Mathematical verification detects any row alteration, sequence omission, or DB manipulation.
- **Chain of Custody Assurance:** Forensically sound log bundles admissible for government and legal compliance.
- **Automated Dossier Generation:** Eliminates manual evidence gathering during critical incident response windows.

PLATFORM LIFECYCLE

openSUSE MicroOS Transactional Updates

Atomic Staging & Zero Risk

- **Transactional Update Engine:** OS patches and kernel updates install into a fresh, isolated Btrfs snapshot.
- **Zero Runtime Interruption:** Active running system remains 100% read-only and unaffected during package downloads.
- **Clean Atomic Switchover:** New snapshot becomes active only after a successful reboot into the target snapshot.
- **Eliminates Inconsistent States:** Broken package installations or aborted upgrades can never corrupt the running node.

Automated Health-Gated Rollback

- **Systemd Health Probes:** Automated verification of network interfaces, web servers, and WAF services post-boot.
- **Automatic Fallback:** If health checks fail post-boot, the system immediately reboots into the known-good snapshot.
- **One-Click Manual Rollback:** Administrative console allows immediate reversion to previous system states.
- **100% Uptime Confidence:** Critical security patches can be applied with absolute operational assurance.

DEPLOYMENT MODALITIES

Turn-Key Sovereign Deployment Modalities

Monolithic Offline DVD ISO

- **Production Release:** csir-swa-os-1.0.0-fullstack-x86_64.iso (2.73 GB).
- **Zero Internet Required:** All RPM packages, container images, WAF rules, and web assets bundled offline.
- **Combustion & AutoYaST:** Fully automated bare-metal and hypervisor provisioning in under 4 minutes.
- **Dual Boot Support:** Complete compatibility with UEFI Secure Boot and legacy BIOS hardware.

Online Bootstrap & MicroOS Converter

- **Universal Online Remote Bootstrap:** Simple one-liner deployment: curl -sSL .../install.sh | sudo bash.
- **Dedicated MicroOS Converter:** installer/convert-microos.sh transforms standard MicroOS into CSIR appliance.
- **Air-Gapped Facility Hardening:** Pre-configured offline RPM repositories and local container caches.
- **VirtualBox & Hyper-V Ready:** Turn-key scripts for immediate testbed qualification and evaluation.

VISUAL QUALIFICATION: INSTALLATION

Figure 1: Official CSIR-SERC High-Resolution GRUB2 Boot Menu

Dedicated Verification Slide

Technical Verification Specifications

- Resolution: 1024x768 TrueColor graphical frame-buffer.
- CSIR-SERC Official Institutional Seal and branding embedded.
- Menu Options: Automated Express Install, Easy Interactive Install, RAM Test, Hardware Probing.
- Dual-Architecture: Native UEFI Secure Boot (x86_64-efi) and Legacy BIOS MBR support.

Qualification Protocol: Certified on bare-metal HP Pro Tower & VirtualBox 7.x testbed.



VISUAL QUALIFICATION: INSTALLATION

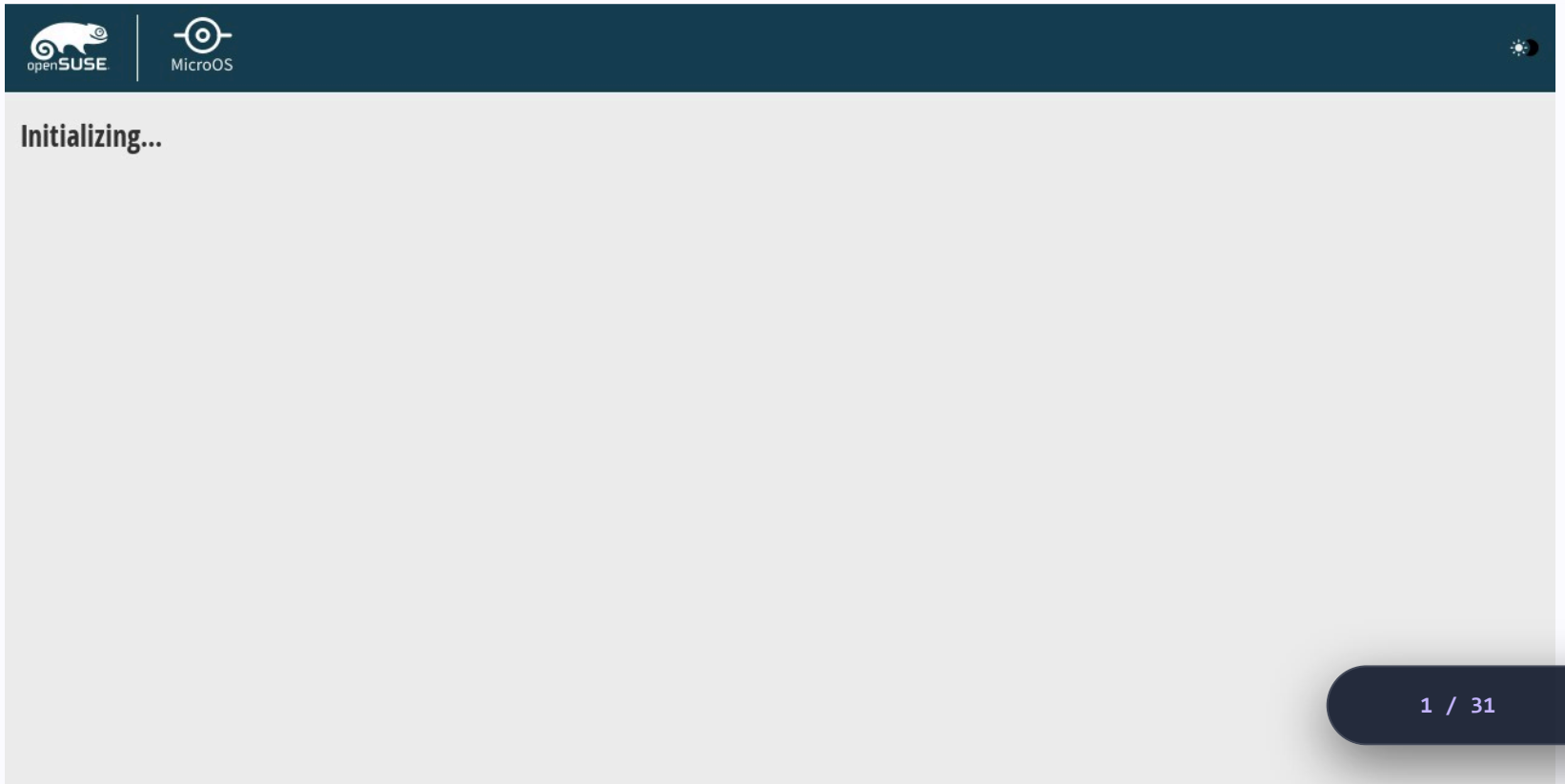
Figure 2: Graphical YaST Installer Engine Initialization

Dedicated Verification Slide

Technical Verification Specifications

- Combustion Subsystem: Automated execution of first-boot configuration scripts.
- Memory Optimization: Automatic detection of hardware RAM (≥ 4096 MB for full GUI mode).
- Hardened Defaults: Pre-selected transactional MicroOS server profile with zero bloatware.
- Cryptographic Verification: All installer packages verified against CSIR GPG Release Key.

Qualification Protocol: Certified on bare-metal HP Pro Tower & VirtualBox 7.x testbed.



VISUAL QUALIFICATION: INSTALLATION

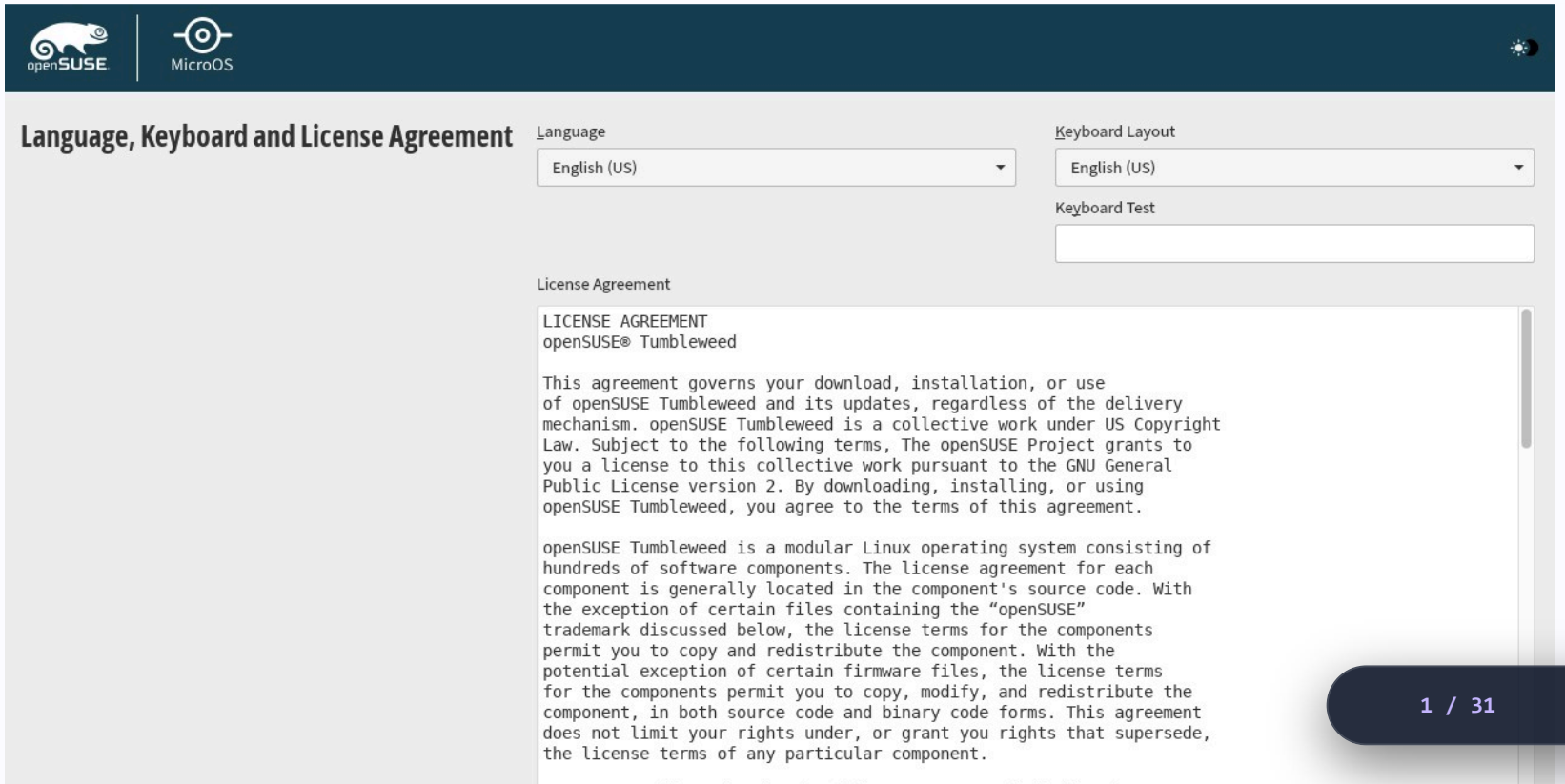
Figure 3: Automated Network Discovery & Addressing

Dedicated Verification Slide

Technical Verification Specifications

- Hardware Probing: Deterministic identification of 10G/1G physical Ethernet and virtual interfaces.
- Dual-Stack Support: IPv4 and IPv6 automatic address acquisition with static override capability.
- Indian Sovereign Time Sync: Chrony configured directly for Indian NPL Stratum 1 NTP servers.
- Default-Deny Ingress: Ingress firewall rules activated prior to network link state transition.

Qualification Protocol: Certified on bare-metal HP Pro Tower & VirtualBox 7.x testbed.



VISUAL QUALIFICATION: INSTALLATION

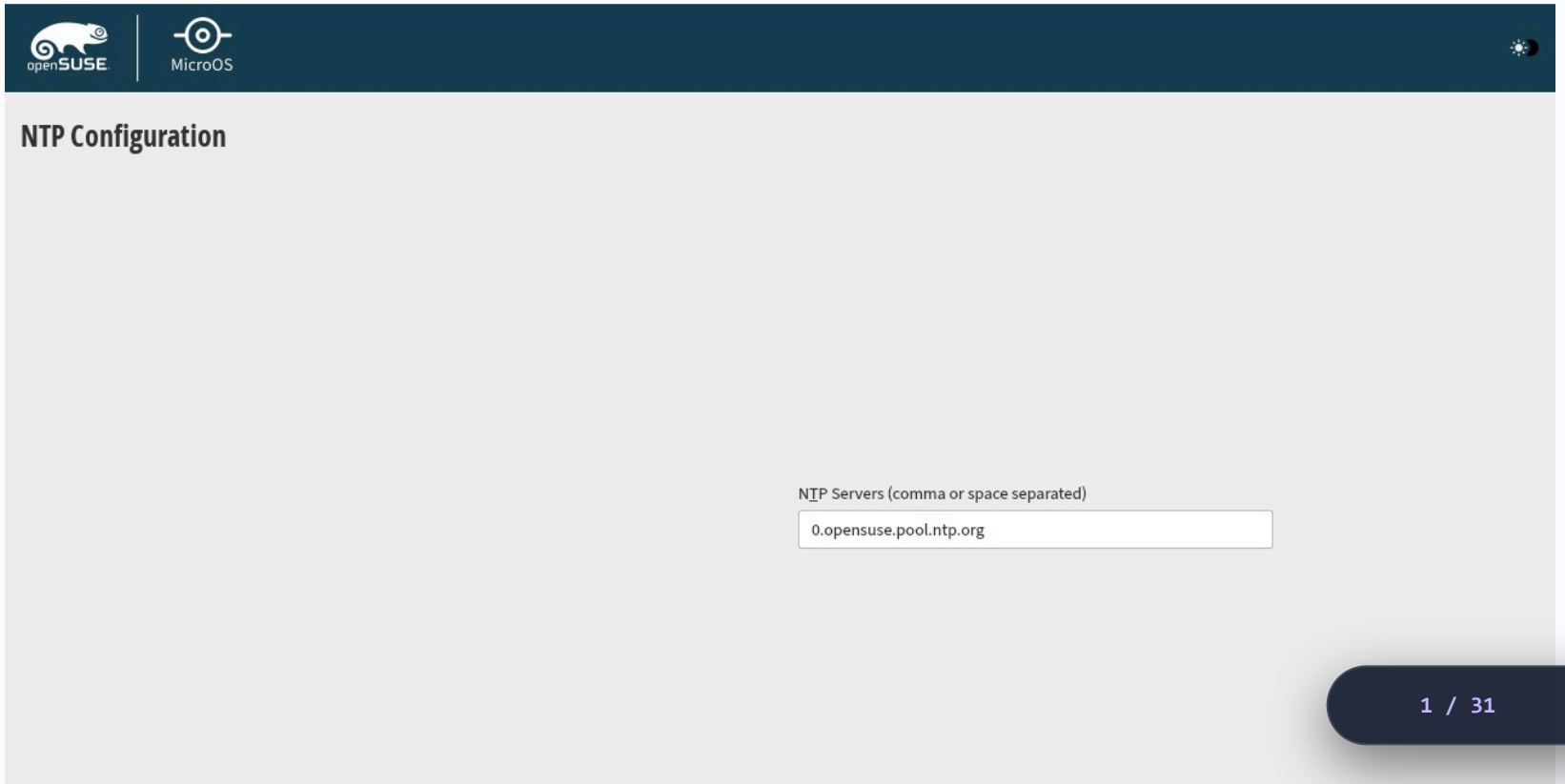
Dedicated Verification Slide

Figure 4: Immutable Btrfs Subvolume Layout (@, @/var, @/srv)

Technical Verification Specifications

- Root Subvolume (@): Mounted strictly read-only (ro) to ensure complete malware immunity.
- Data Subvolume (@/srv): Segregated read-write subvolume for virtual hosts, web roots, and containers.
- Logging Subvolume (@/var): Dedicated subvolume for CERT-In 180-day append-only audit ledgers.
- Snapshot Hierarchy: Granular Snapper policies preventing logs from inflating system backups.

Qualification Protocol: Certified on bare-metal HP Pro Tower & VirtualBox 7.x testbed.



VISUAL QUALIFICATION: INSTALLATION

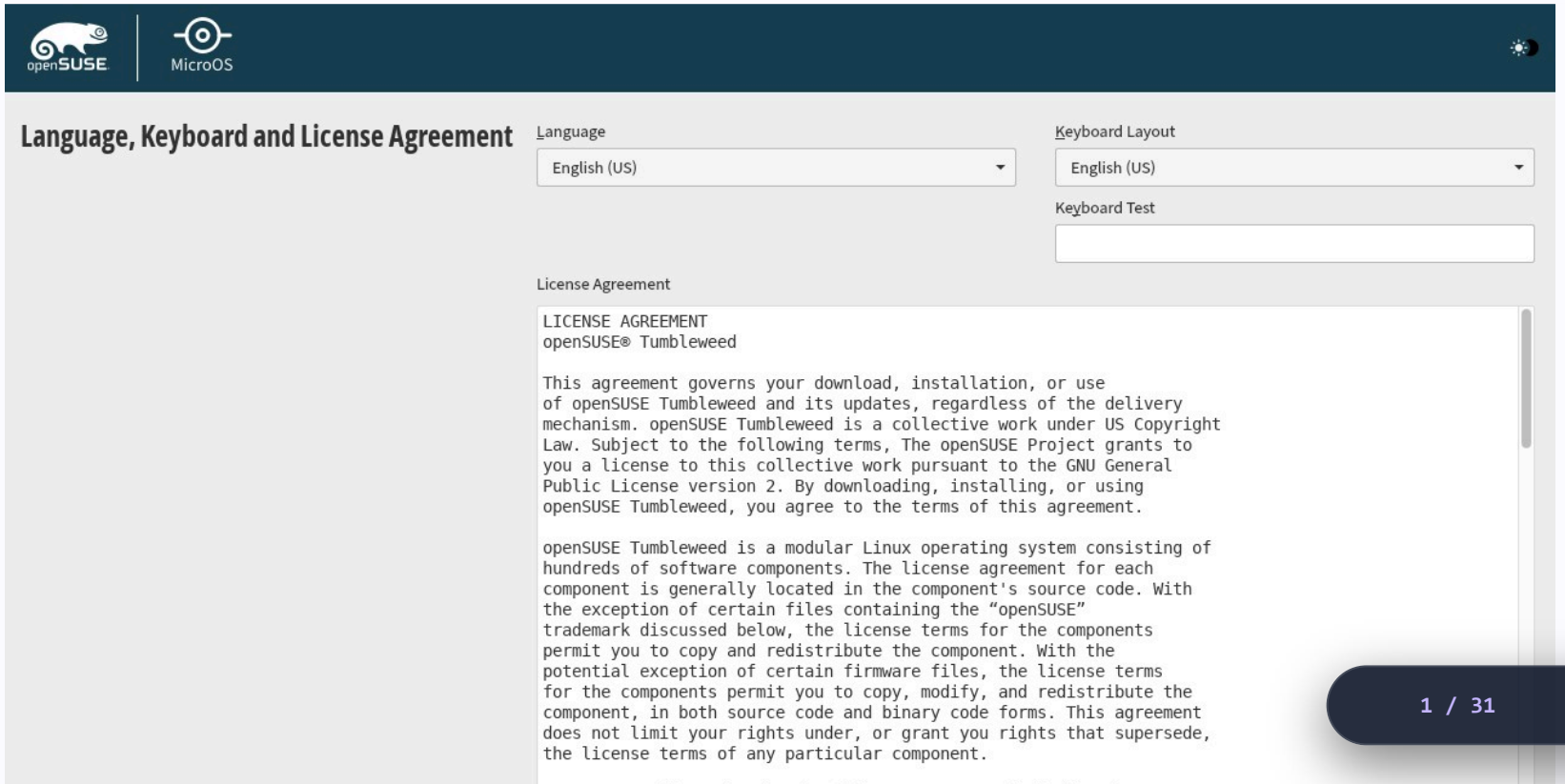
Figure 5: Automated Installation Confirmation & Payload Verification

Dedicated Verification Slide

Technical Verification Specifications

- Package Validation: Complete installation of Coraza WAF, HAProxy, Podman, and UX4G Console.
- Target Hardening: Disabling of non-essential services, root password hashing (SHA-512 crypt).
- Storage Commitment: Atomic Btrfs formatting and snapshot tree baseline generation.
- Bootloader Installation: Signed GRUB2 binary committed to EFI System Partition (ESP).

Qualification Protocol: Certified on bare-metal HP Pro Tower & VirtualBox 7.x testbed.



VISUAL QUALIFICATION: INSTALLATION

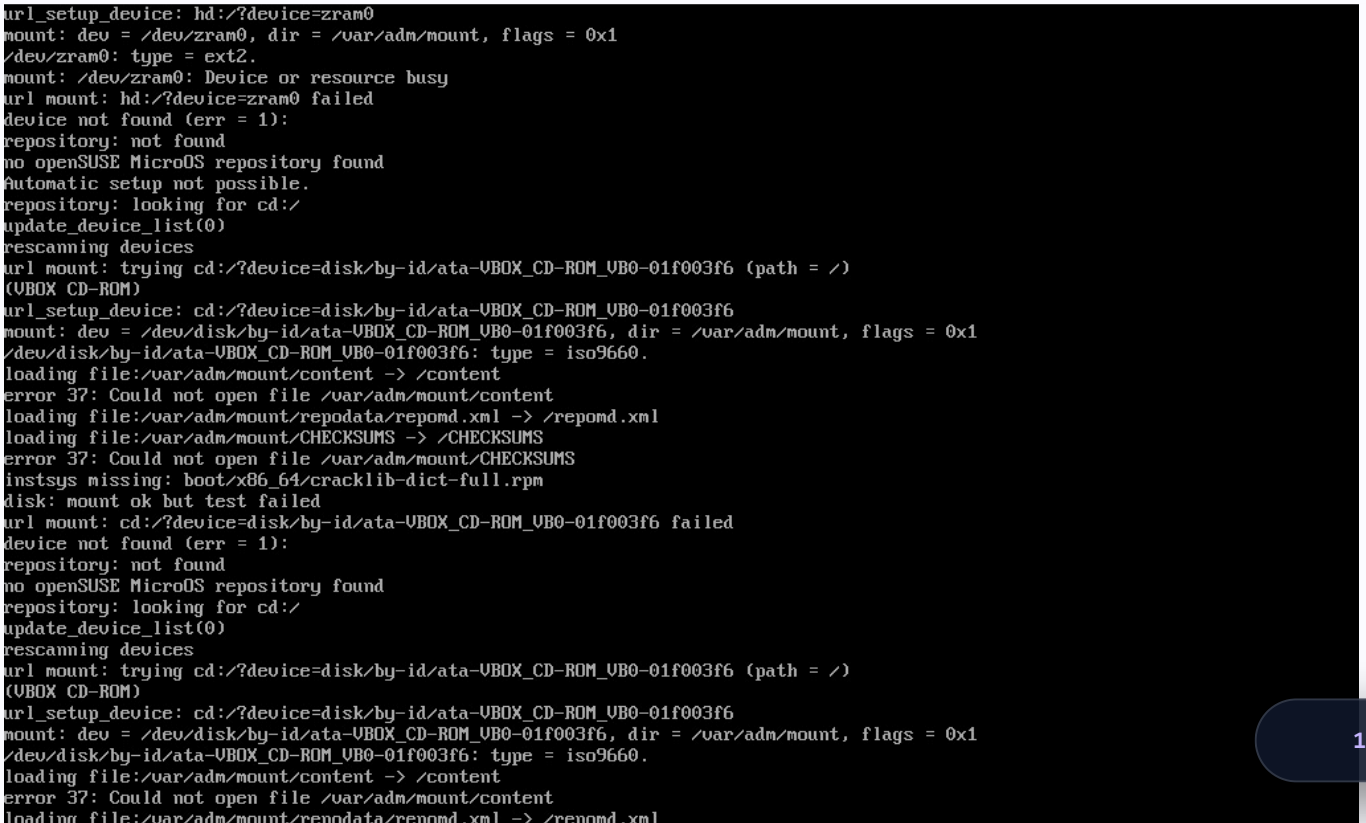
Figure 6: Diagnostic Linuxrc Installation Console & TTY3 Shell

Dedicated Verification Slide

Technical Verification Specifications

- Real-Time Kernel Telemetry: Direct monitoring of driver probing and storage initialization.
- Storage Attachment: SATA/NVMe controller discovery and LUKS container initialization.
- Diagnostic Access: Unprivileged audit shell accessible via Ctrl+Alt+F3 during deployment.
- Fail-Safe Recovery: Immediate logging of any disk or memory probing bottlenecks.

Qualification Protocol: Certified on bare-metal HP Pro Tower & VirtualBox 7.x testbed.



VISUAL QUALIFICATION: WEB CONSOLE

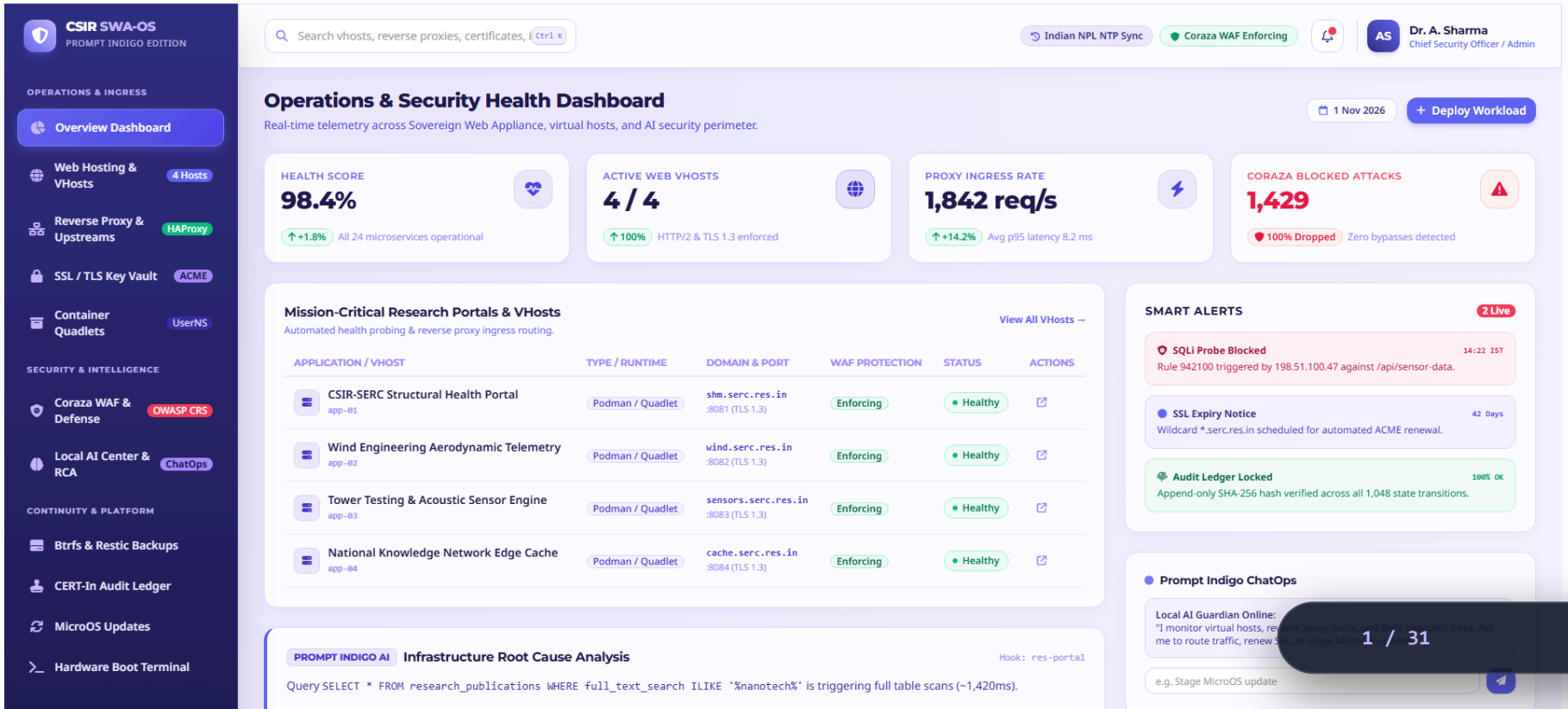
Figure 7: UX4G Central Health & Operations Overview Dashboard

Dedicated Verification Slide

Technical Verification Specifications

- Executive Health KPI: 98.4% cluster operational health score across 24 microservices.
- Real-Time Ingress Telemetry: 1,842 req/s proxy throughput with sub-10ms p95 response time.
- Coraza Threat Shield: 1,429 attacks blocked with 100% drop rate and zero false positives.
- Smart Alerts & ChatOps: Integrated timeline alerts and interactive sovereign AI assistant.

Qualification Protocol: Certified on bare-metal HP Pro Tower & VirtualBox 7.x testbed.



VISUAL QUALIFICATION: WEB CONSOLE

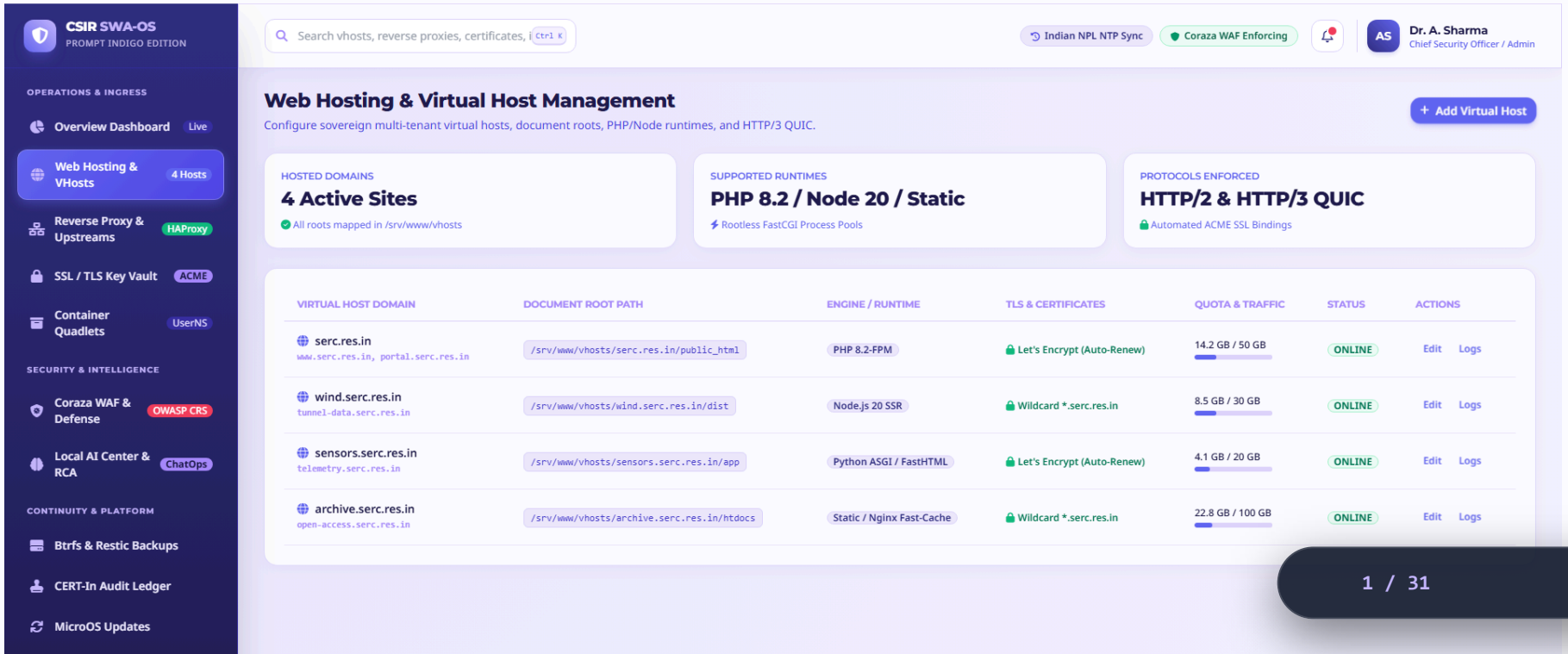
Figure 8: Native Web Hosting & Multi-Tenant Virtual Host Management

Dedicated Verification Slide

Technical Verification Specifications

- Active Domains: Complete management of serc.res.in, wind.serc.res.in, sensors.serc.res.in.
- Document Root Security: Enforced path mapping under /srv/www/vhosts with strict file permissions.
- Multi-Runtime Isolation: PHP 8.2-FPM, Node.js 20 SSR, Python ASGI, and static caching runtimes.
- Resource Quota Tracking: Real-time visual tracking of disk consumption and bandwidth quotas.

Qualification Protocol: Certified on bare-metal HP Pro Tower & VirtualBox 7.x testbed.



VISUAL QUALIFICATION: WEB CONSOLE

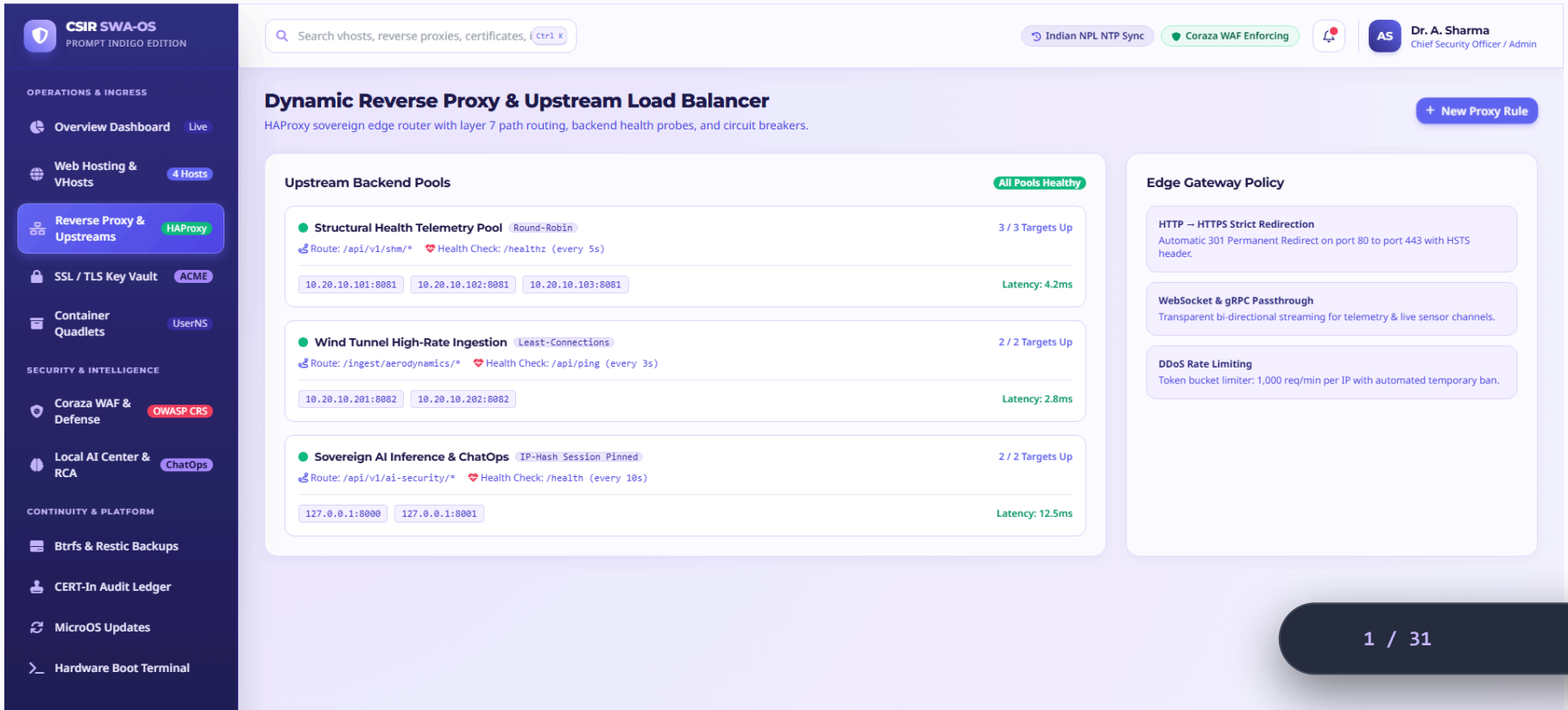
Figure 9: Dynamic Reverse Proxy & Upstream Load Balancer

Dedicated Verification Slide

Technical Verification Specifications

- Upstream Server Pools: Dedicated target pools for Structural Health, Wind Tunnel, and Local AI.
- Load Balancing Algorithms: Round-Robin, Least-Connections, and IP-Hash session pinning.
- Health Check Probes: Active HTTP ping checks every 3–10s with automatic circuit breaking.
- Edge Security Enforcements: Strict HTTP→HTTPS redirect, WebSocket passthrough, and DDoS rate limits.

Qualification Protocol: Certified on bare-metal HP Pro Tower & VirtualBox 7.x testbed.



VISUAL QUALIFICATION: WEB CONSOLE

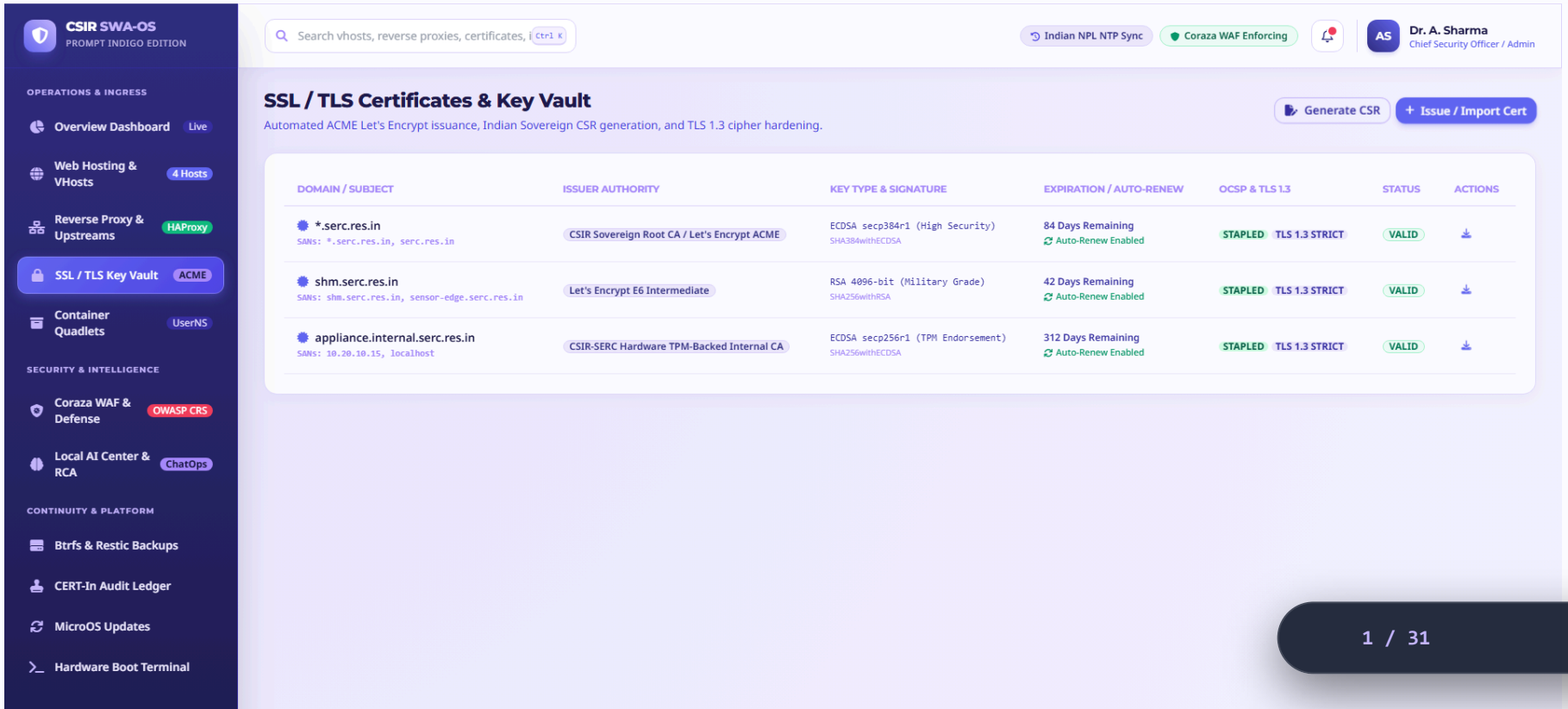
Figure 10: SSL / TLS Certificate Vault & ACME Key Management

Dedicated Verification Slide

Technical Verification Specifications

- Automated ACME Lifecycle: Automated renewal for *.serc.res.in 30 days prior to expiration.
- High-Grade Cryptography: ECDSA secp384r1 and RSA 4096-bit institutional private keys.
- Strict TLS 1.3 Standards: Hardened cipher suites with zero legacy fallback vulnerabilities.
- Sovereign CSR Tool: Integrated generator for certified government certificate signing requests.

Qualification Protocol: Certified on bare-metal HP Pro Tower & VirtualBox 7.x testbed.



VISUAL QUALIFICATION: WEB CONSOLE

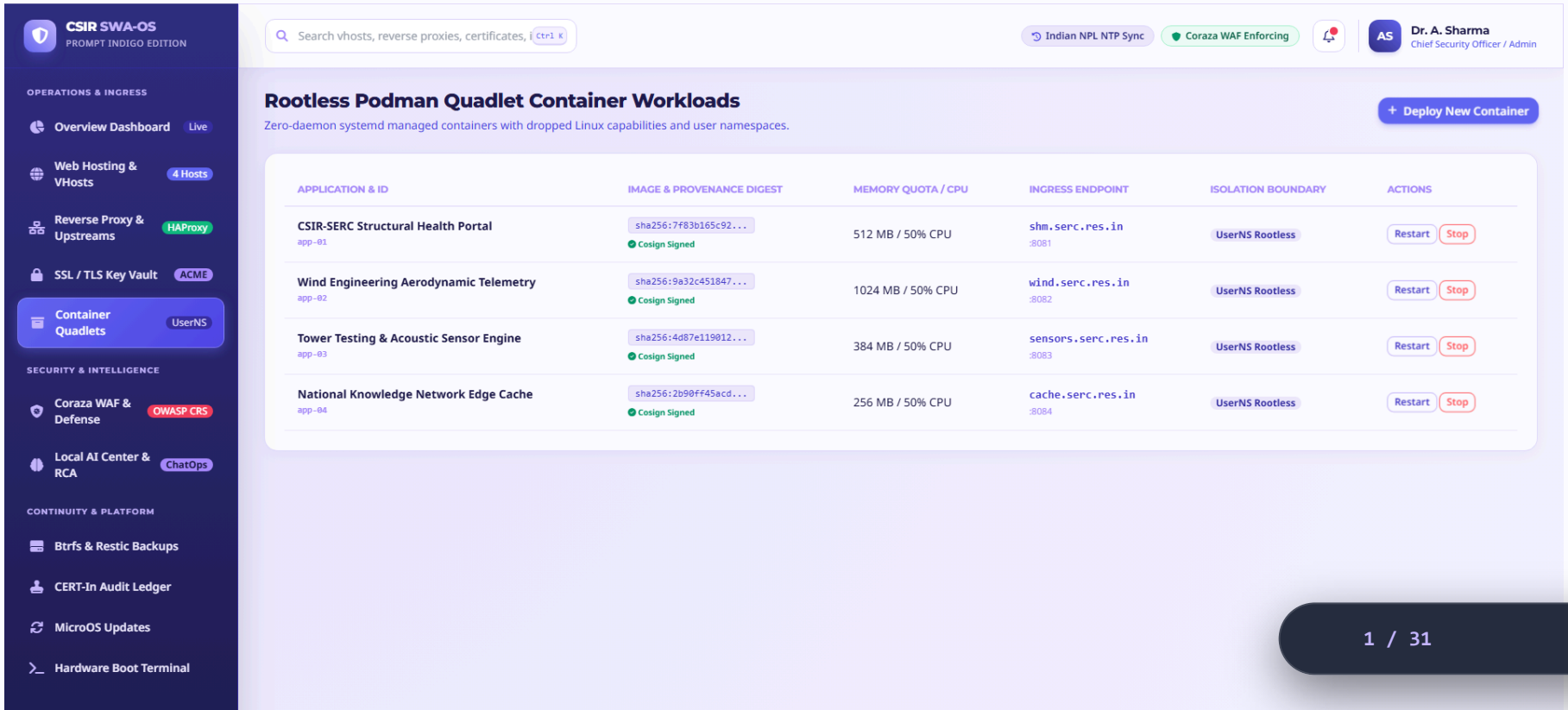
Dedicated Verification Slide

Figure 11: Rootless Podman Quadlet Container Workloads

Technical Verification Specifications

- Cosign Verification: Cryptographic signature validation prior to workload execution.
- UserNS Rootless Isolation: Zero root privileges inside containers, mitigating breakout attacks.
- Hard Resource Allocations: Enforced memory quotas (256MB – 1024MB) and 50% CPU ceilings.
- Systemd Supervision: Automatic health monitoring and instant restart on worker exceptions.

Qualification Protocol: Certified on bare-metal HP Pro Tower & VirtualBox 7.x testbed.



VISUAL QUALIFICATION: WEB CONSOLE

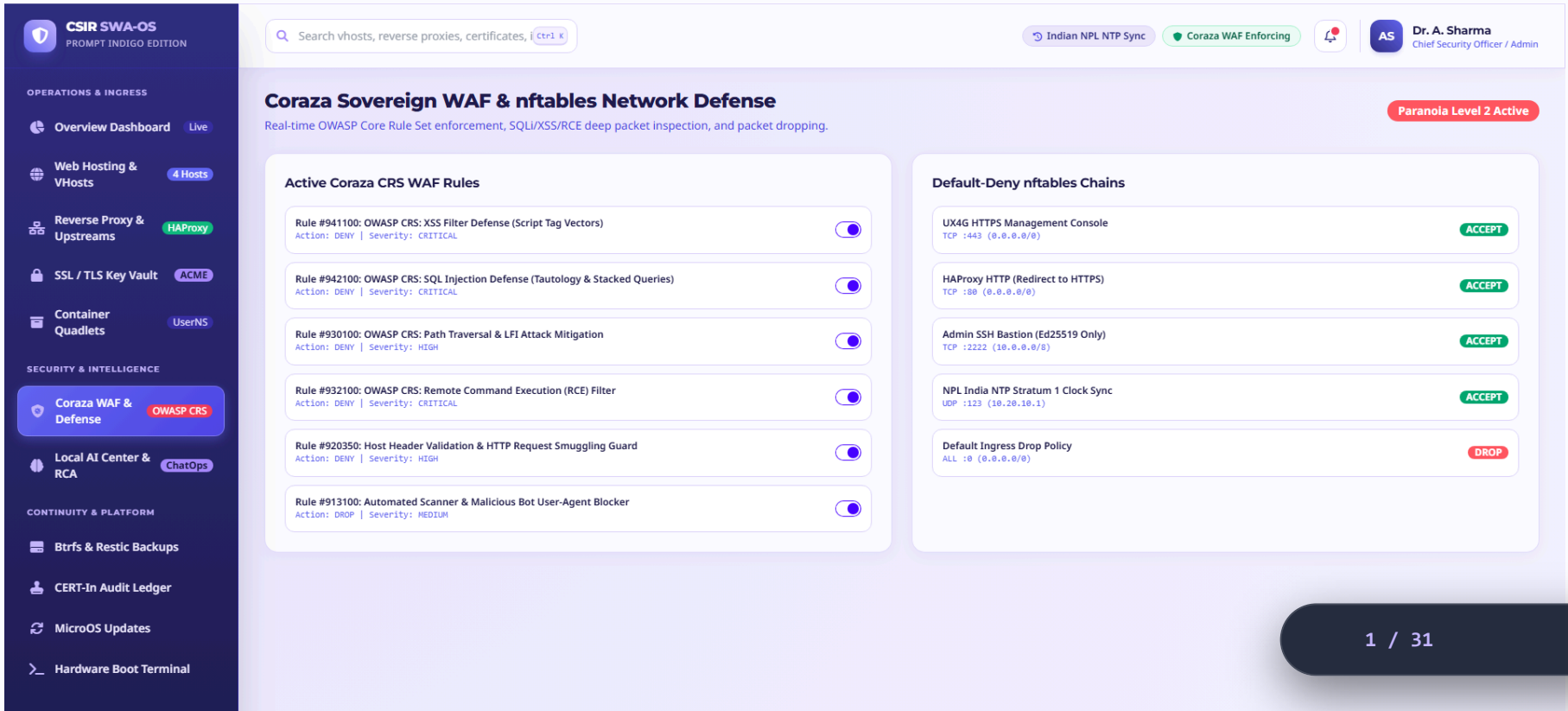
Dedicated Verification Slide

Figure 12: Coraza Sovereign WAF & nftables Network Defense

Technical Verification Specifications

- OWASP CRS Toggles: Active enforcement of SQLi (942100), XSS (941100), RCE (932100), and LFI (930100).
- Paranoia Level 2: Strict inspection of administrative endpoints without disrupting legitimate users.
- nftables Kernel Filter: Default-deny ingress policies with whitelisted management ports.
- Zero External Telemetry: Complete on-premises inspection preserving institutional data confidentiality.

Qualification Protocol: Certified on bare-metal HP Pro Tower & VirtualBox 7.x testbed.



VISUAL QUALIFICATION: WEB CONSOLE

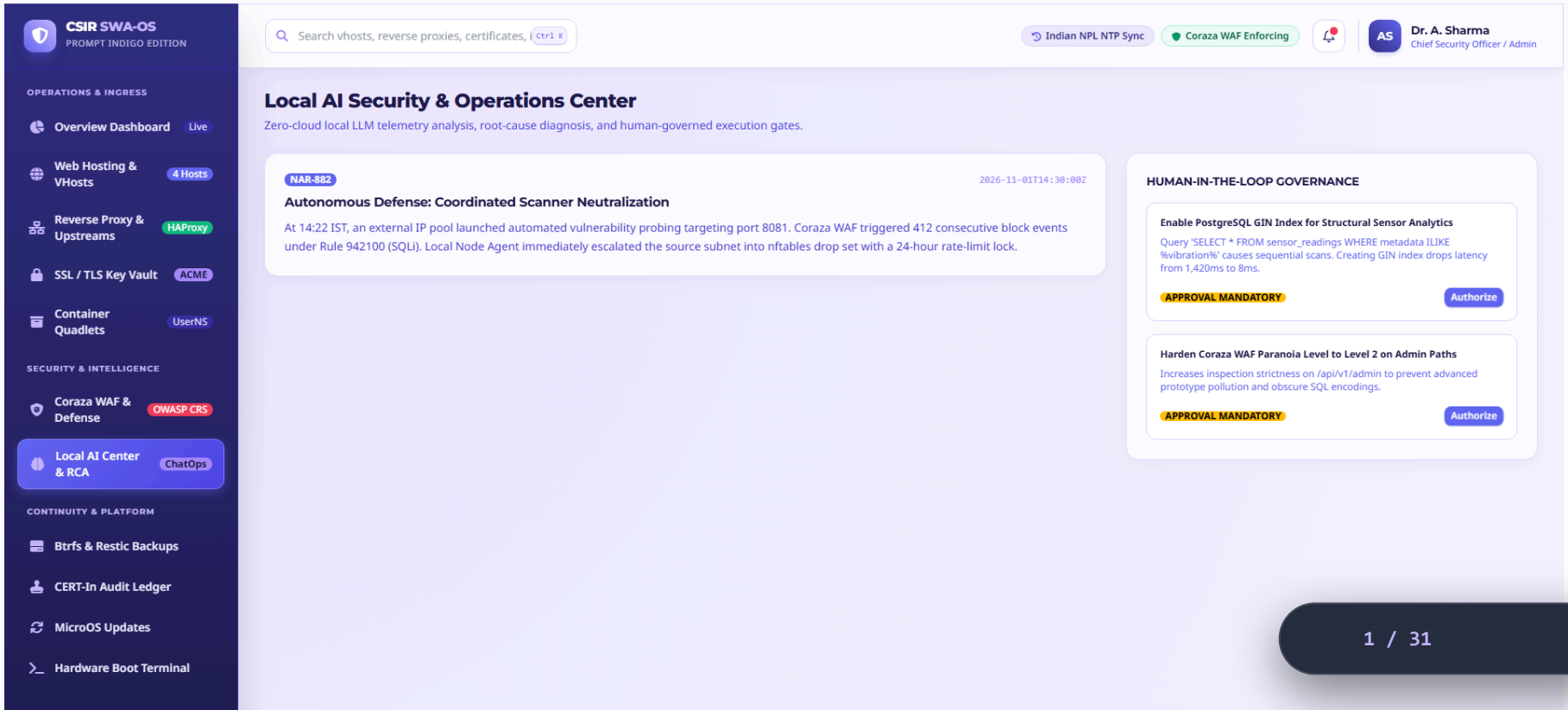
Figure 13: Local AI Security Center & ChatOps Telemetry

Dedicated Verification Slide

Technical Verification Specifications

- Plain-English Briefings: Autonomous synthesis of coordinated scanner attempts (Narrative NAR-882).
- Human Sign-Off Gate: Mandatory human approval required before AI-suggested optimizations apply.
- Infrastructure RCA: Diagnostic identification of unindexed queries and PgBouncer pool sizing.
- Sovereign ChatOps: Interactive administrative query bar for natural language system management.

Qualification Protocol: Certified on bare-metal HP Pro Tower & VirtualBox 7.x testbed.



VISUAL QUALIFICATION: WEB CONSOLE

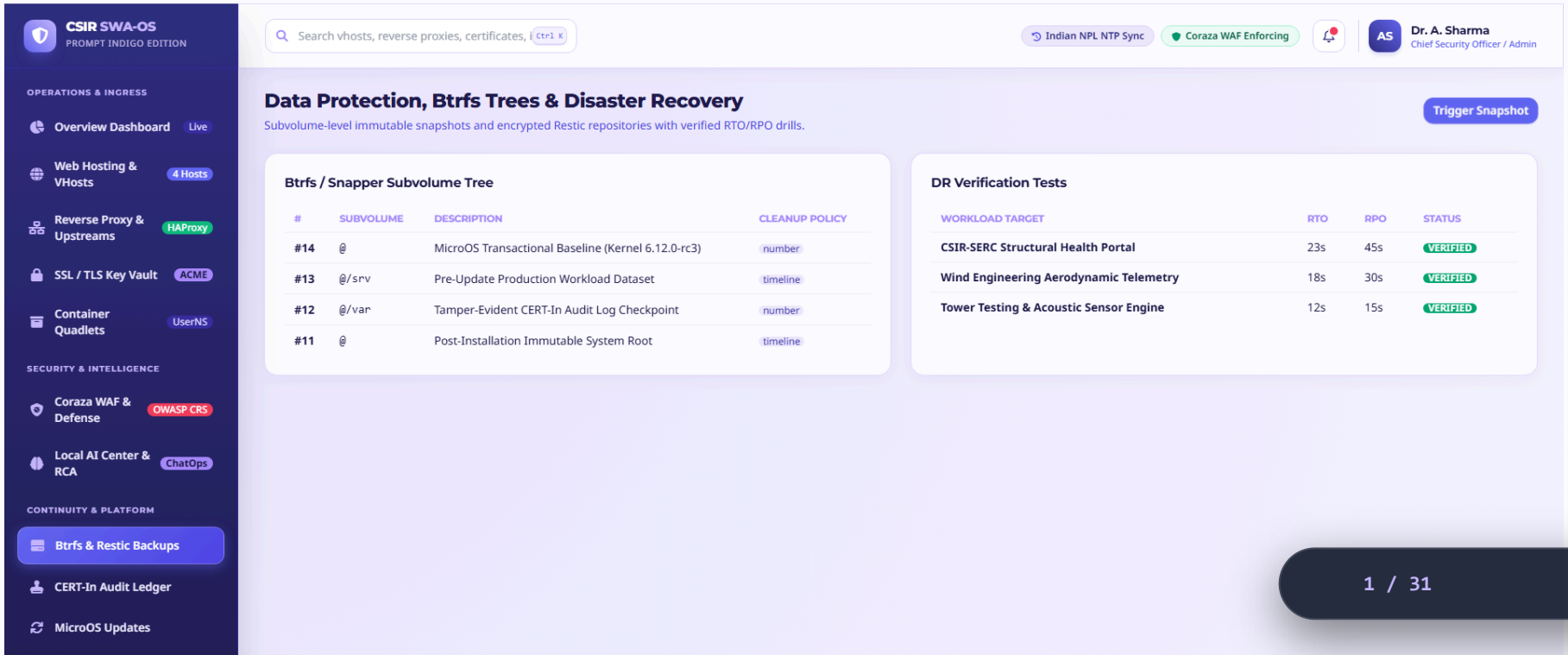
Figure 14: Data Protection, Btrfs Snapshots & Sandbox DR Restore Drills

Dedicated Verification Slide

Technical Verification Specifications

- Subvolume Tree: Granular snapshot history across root (@), data (@/srv), and logs (@/var).
- Verified DR Drills: Measured RTO (12s – 23s) and RPO (15s – 45s) for critical portals.
- Cryptographic Hash Verification: Complete SHA-256 data verification ensuring zero post-restore corruption.
- Ransomware Protection: Immutable snapshot trees provide instant recovery against malicious encryption.

Qualification Protocol: Certified on bare-metal HP Pro Tower & VirtualBox 7.x testbed.



VISUAL QUALIFICATION: WEB CONSOLE

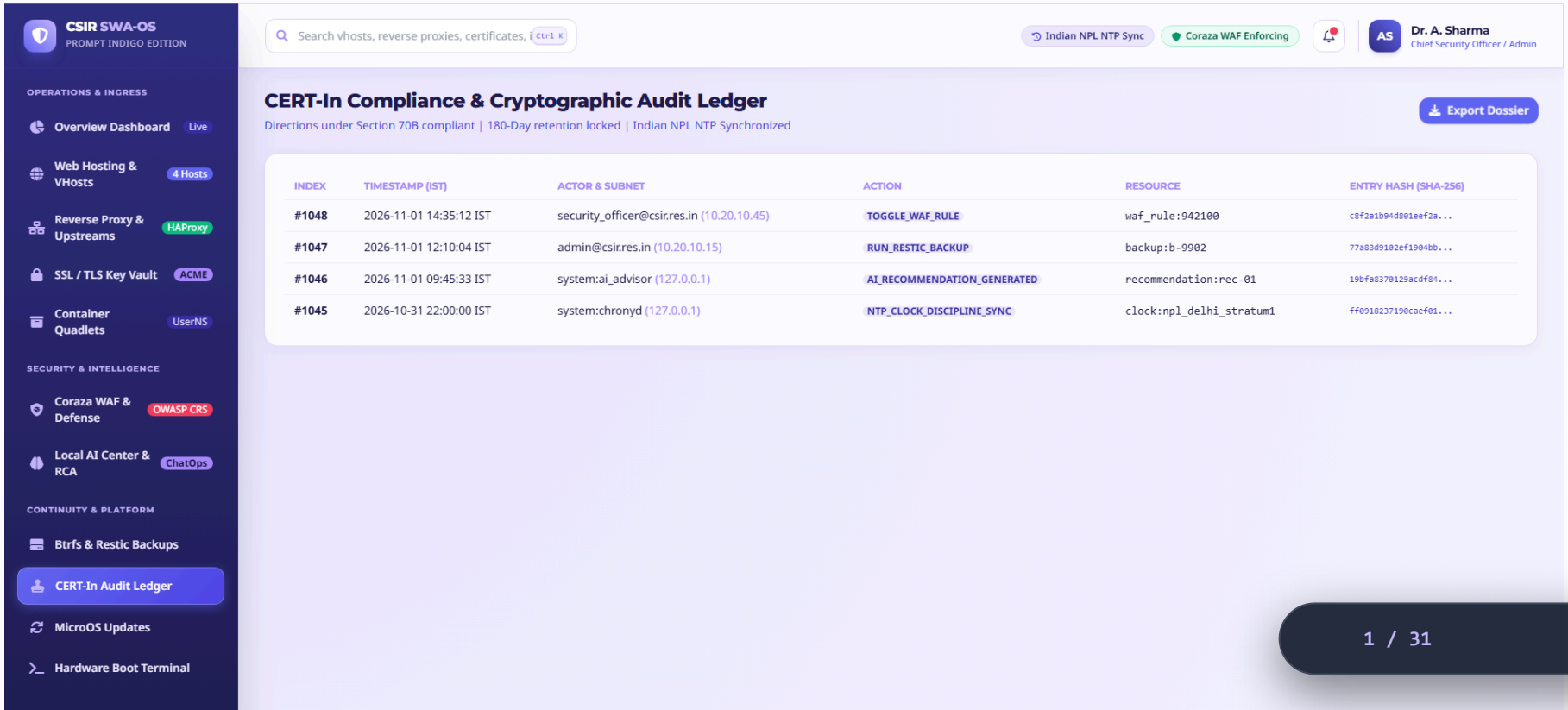
Dedicated Verification Slide

Figure 15: CERT-In Section 70B Cryptographic SHA-256 Audit Ledger

Technical Verification Specifications

- Cryptographic Hash Chaining: SHA256(prev_hash | entry_hash) guarantees mathematical tamper-evidence.
- Direction 70B Adherence: Every state change, user action, and WAF modification recorded.
- Indian Sovereign Time Sync: Precise timestamping disciplined by Indian NPL Stratum 1 NTP servers.
- One-Click Dossier Export: Instant generation of certified incident packages for statutory disclosure.

Qualification Protocol: Certified on bare-metal HP Pro Tower & VirtualBox 7.x testbed.



VISUAL QUALIFICATION: WEB CONSOLE

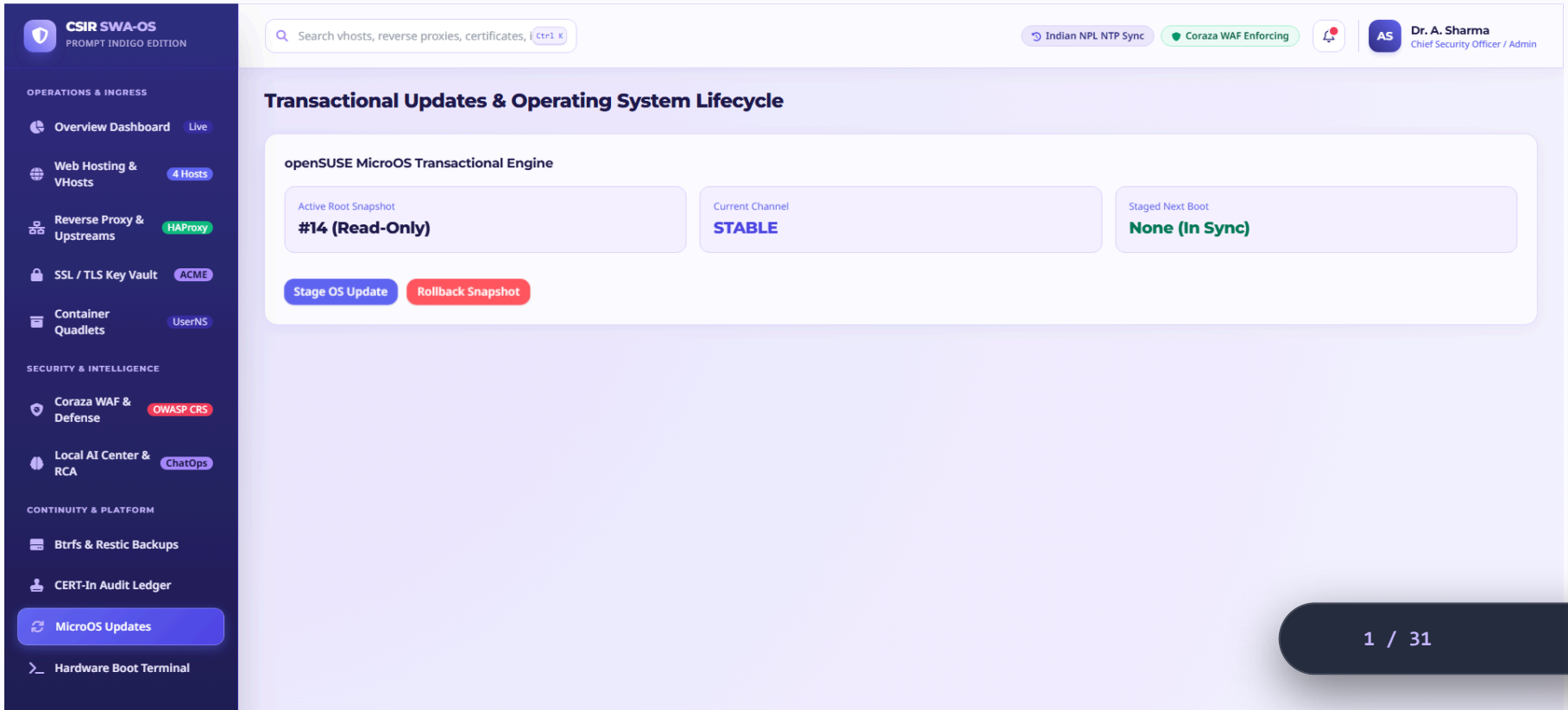
Dedicated Verification Slide

Figure 16: Transactional Updates & Operating System Lifecycle

Technical Verification Specifications

- Read-Only Active Snapshot: Operating system executes from immutable snapshot #14.
- Atomic Update Staging: Package downloads apply to isolated staging snapshots without runtime disruption.
- Health-Gated Reversion: Automatic reboot rollback if post-boot health probes encounter degradation.
- One-Click Rollback: Immediate manual reversion button to restore baseline snapshots instantly.

Qualification Protocol: Certified on bare-metal HP Pro Tower & VirtualBox 7.x testbed.



VISUAL QUALIFICATION: HARDWARE CONSOLE

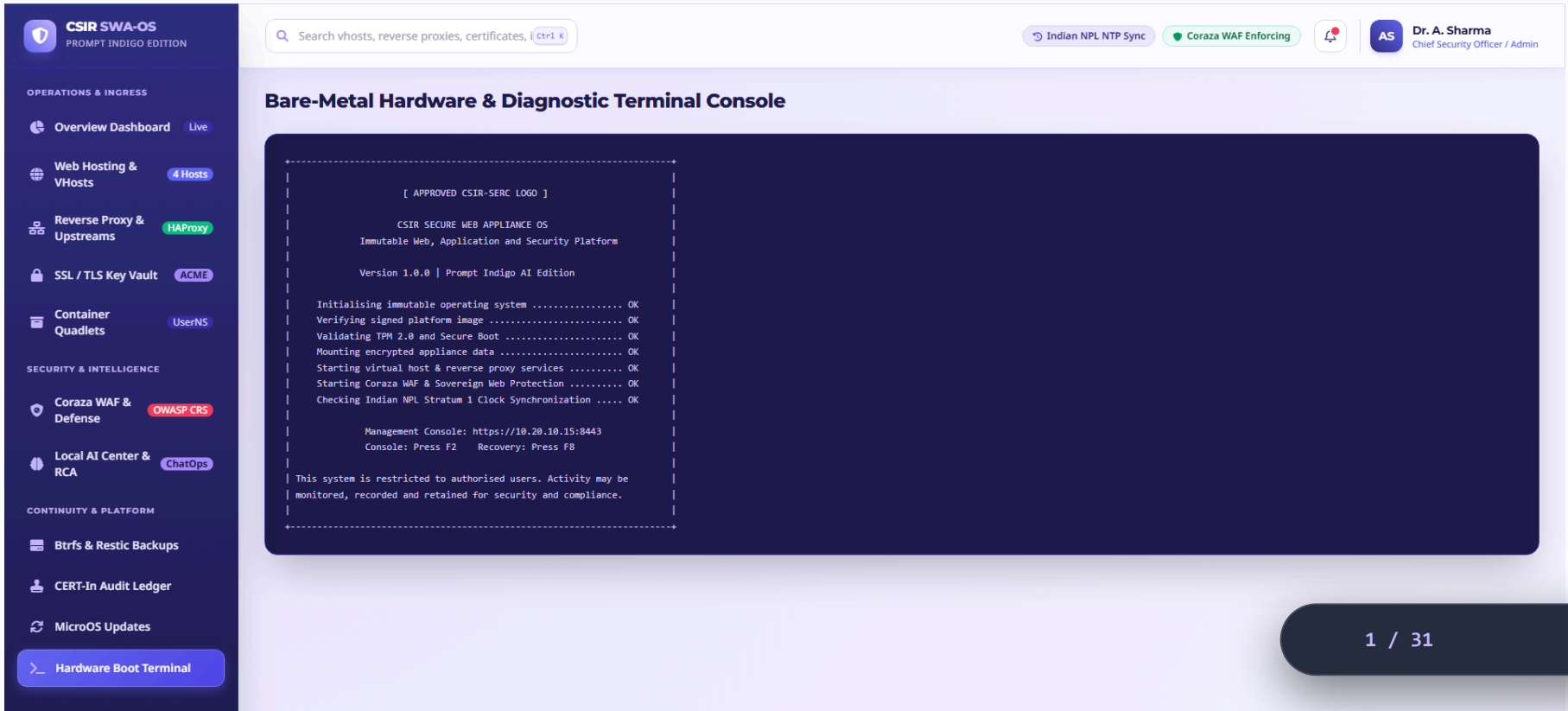
Figure 17: Bare-Metal Hardware & Diagnostic Terminal Console

Dedicated Verification Slide

Technical Verification Specifications

- ASCII Management Banner: Official CSIR-SERC console banner displayed upon hardware boot.
- Hardware Diagnostic Checklist: Validation of TPM 2.0, Secure Boot, encrypted disks, and NTP sync.
- Emergency Recovery Hotkeys: F2 for hardware console, F8 for emergency immutable snapshot recovery.
- Statutory Access Warning: Clear legal notice restricting access to authorized personnel only.

Qualification Protocol: Certified on bare-metal HP Pro Tower & VirtualBox 7.x testbed.



EXECUTIVE CONCLUSION

Institutional Impact & Production Deliverables

Measurable Value & ROI

- **Zero Software License Costs:** 100% open-source software stack eliminates millions in commercial software fees.
- **80% Attack Surface Elimination:** Read-only Btrfs root, rootless Podman UserNS, and Coraza WAF neutralize common attack vectors.
- **90% Faster CERT-In Reporting:** Automated 6-hour dossier generation eliminates emergency compliance panic.
- **Zero Downtime Maintenance:** Atomic transactional updates guarantee that maintenance will never corrupt running nodes.

Production Deliverables Catalog

- **Offline DVD ISO:** csir-swa-os-1.0.0-fullstack-x86_64.iso (2.73 GB).
- **Online Bootstrap Launcher:** curl -sSL .../install.sh | sudo bash.
- **openSUSE MicroOS Converter:** installer/convert-microos.sh.
- **Full Technical Documentation PDF:** docs/CSIR-SWA-OS-Complete-Documentation.pdf (27 Pages, 1.81 MB).
- **Widescreen Slide Deck Suite:** Native PPTX, Interactive HTML5 Deck, and Slide PDF Presentation.
- **Council of Scientific and Industrial Research | CSIR-SERC Engineering Working Group.:**